8 Discrete Input & Output Devices and General Manufacturing Measurements

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Topic Highlights

Actuation Technologies and Their Control
Sensing Technologies and Interfacing Techniques
Remote and Networked I/O

8.1 Introduction

Historically, the practice of discrete control evolved quite differently from continuous process control, with very little overlap in technologies and applied principles. Early discrete manufacturing machines were wholly mechanical contrivances, harnessing waterpower through ingenious combinations of drive shafts, cams, and levers to accomplish ever more complex automation goals.

The advent of electrical motors caused little change at first, with mechanical control still the norm. Highly automated assembly and fabrication systems were often based on a central camshaft driven by a motor, with the entire sequence of the system defined by a single turn of the camshaft. All motions were driven by cams on this shaft, with lobes triggering activity at the appropriate interval.

8.1.1 Separating the Control Function

The first real separation of control and actuation came with the increasing use of electromechanical relays, especially in the automotive industry. This ultimately led to complex control configurations of these relays arranged to create latches and combinatorial functions, creating motion through the use of electrically driven hydraulic or pneumatic valves (solenoid-actuated valves or, simply *solenoid valves*).

In this new world of separate control and actuation, some functionality was actually lost. The actuation technologies were crude and simple: pneumatic and hydraulic cylinders which, when the related solenoid valve was triggered, would dumbly extend to the limit of their travel as defined by a mechanical stop. Speed of travel of these actuators could be controlled to some degree by installing a flow valve or pressure regulator in the hydraulic or pneumatic supply line, and these could be adjusted manually.

But the exacting control of acceleration and deceleration through the entirety of the stroke, which had been possible by skilled practitioners of cam technology, had been lost and would not be regained until new actuation technologies were introduced.

Beginning in some industries as early as the 1980s, additional sophistication of actuation and control became increasingly necessary. Compelled by the need for further automation spurred by increasing labor costs, as well as by increasing demands among consumers for broad product choices and custom-

ization, automation practitioners were required to produce more flexible machines. Often these machines were required to be soft-configurable—automatically adjustable under electronic control to manufacture slightly different products, or products of different sizes.

This trend led to the use of more flexible actuation technologies, including *servo motors*, *stepping motors*, and servo-controlled hydraulics. With corresponding advances in control, these technologies regained the precise control over motion, adding the benefit of automatic adjustment under software control.

8.1.2 Evolution in Sensing Technologies

A simultaneous trend dramatically affected the sensing technologies used in discrete control. In early systems, *limit switches* were often employed to sense the end of travel of mechanical motions driven by simple pneumatic cylinders. Often these limit switches were incorporated directly into a control scheme to trigger the next function in sequence.

Flexible machinery, however, brought with it the requirement for more flexible sensing. In many cases it was no longer adequate to sense when an actuator reached a single position; rather, it became necessary to sense multiple positions, or to continuously sense the absolute position of an actuator at all times. Sensing devices such as *encoders*, *linear potentiometers*, and *magnetostrictive transducers* came into more widespread use to accomplish this goal.

Sensing technology has also been driven by a parallel trend—that toward the imposition of ever-greater quality requirements, with the corresponding need to measure and control all relevant parameters relating to the manufacture of a product. It is now common for discrete manufacturing equipment to measure temperatures and hydraulic or pneumatic pressures present at the time of a product's manufacture, as well as to take critical dimensional measurements of a product either in-process or upon completion. This data is then used either in a control scheme to meet qualitative goals, or is stored as proof of product quality.

Therefore, where previously *digital inputs* (those capable of sensing only "on" or "off" states) were employed in discrete control, now *analog inputs* are increasing in widespread use to sense the many variables of interest to the control engineer.

The rest of this topic will look at the technologies and devices mentioned above in greater detail.

8.2 Actuation Technologies and Their Control

Actuation technologies in common use today include the following:

- Fluid power devices
    - Pneumatic cylinders, rotary actuators, and air motors
    - Hydraulic cylinders and motors
- Electrical motors
    - DC motors
    - AC synchronous motors and variable frequency drives
    - Servo motors
    - Stepping motors
    - There are also a number of specialty actuators in use, such as *piezoconstrictive devices* and *voicecoil actuators*, which will be described in passing.

The choice of an actuation technology is typically dictated by the characteristics of each technology and how those characteristics match the needs and circumstances of an application. For example:

Pneumatic cylinders can create linear motion with reasonably high degrees of force in a small space. The amount of force achievable is determined by the air pressure applied multiplied by the area of the cylinder's piston, as shown in Equation 8-1: Force Obtained from a Cylinder Actuator.

$$F = PA \tag{8-1}$$

where:

F = Force in pounds

P = Applied pressure in PSI

A = Area of cylinder in square inches

With adequate air supply, high actuation speeds are also possible, making pneumatic actuation suitable for many small, high-speed automation systems, and indeed the technology is in widespread use because of its advantages.

In applications requiring great levels of force, however, two limitations of pneumatic actuation become prominent. The first is the inherent compressibility of the actuation medium, air. This implies that there will be at least a small amount of resilience present in a pneumatic system.

The second limitation is the pressure available for actuation. For reasons of safety and practicality, air pressure as distributed in typical "shop air" systems is limited to roughly 80 PSI (5.5 bar). For a given cylinder diameter, this puts an upper limit on the amount of force achievable.

For higher force applications, hydraulic systems are often employed with pressures up to 5000 PSI (345 bar). Disadvantages include cost, since localized hydraulic pumps are often required, and the inevitable leakage of hydraulic fluid, which precludes use of the technology in clean environments.

As the cost of related control technology declines, increasing use is being made of electrical actuation technologies, including servo motors and stepping motors. With appropriate gearing, these devices can generate great amounts of force, although the force is typically in the form of rotary motion. To obtain linear motion, which is often the requirement in automated systems, a *lead screw* or belt drive is frequently employed. Drive and control costs, as well as the cost of conversion of rotary to linear motion, can be a substantial impediment to using electrical motor actuation, but the cleanliness and extent of control that are possible are often overriding factors.

8.2.1 Control of Fluid Power Actuation

The most common method of interfacing fluid power (hydraulic or pneumatic) actuators to a control system is through the use of *solenoid valves*. These are valves which are actuated by means of an electric solenoid—application of electrical power to a coil of wire generates a magnetic field, which acts upon a metallic actuator that switches the valve.

8.2.2 Configurations of Solenoid Valves

Pneumatic valves for control purposes are often categorized as two-way, three-way or four-way valves. A two-way valve possesses two ports to which pneumatic connections may be made. When actuated, the valve creates a connection between the two ports (assuming "normally closed" design of the valve) allowing flow to take place. Such a valve might be used in an air motor, for example, where air flow is applied to make the motor turn, and removed to make it come to a stop.

Three-way valves are outfitted with three ports. One of these (the "load" port) is switched between a connection to the "exhaust" port and, when actuated, the "supply" port. Such valves are often used to control *single-acting* pneumatic cylinders. These are cylinders driven by compressed air in only one direction, and are outfitted with a spring that will return them to a rest position. A source of compressed air is connected to the supply port, and the cylinder is connected to the load port. Upon actuation, these two ports are interconnected, allowing compressed air to flow to the cylinder, which

overcomes its spring and extends. Upon deactivation, however, the pressure behind the cylinder's piston must be relieved or the cylinder will fail to retract. This is accomplished by leaving the exhaust port of the valve open to atmosphere. When the valve returns to its deactivated position, the load port (connected to the pressurized cylinder) is connected back to the exhaust port, allowing the pressure to release to atmosphere.

Four-way valves are typically used to control *double-acting* cylinders—those having pneumatic connections on either side of the piston. Two valve connections are provided for exhaust and supply, and two more for the two ends of the cylinder under control. In one position, the valve connects one end of the cylinder to the supply pressure and the other to exhaust. When the valve is switched, these connections are reversed.

Hydraulic valves work similarly, except that instead of exhausting compressed air to atmosphere, depressurized hydraulic fluid is returned via plumbing to a reservoir tank.

Miniature solenoid valves intended for small automated machines can be quite small—as little as 0.5 cu. in. (8 cc)—and are often mounted on air manifolds that provide a common air supply to the valves without individual plumbing. Some suppliers provide valve assemblies mounted on printed circuit boards, simplifying both pneumatic and electrical connection of the devices.

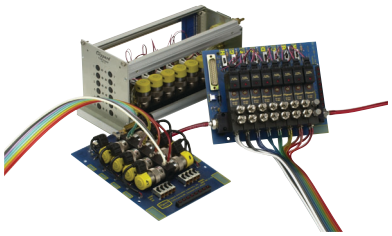


Figure 8-1: Valve Manifolds Integrated with Electronic Circuit Boards (Courtesy: Clippard Instrument Laboratory, Inc.)

Where greater air or hydraulic flows are required than are possible with small low-power valves, two options exist. The first is obvious: use a larger valve, switched by a higher-power solenoid, often requiring additional electrical interfacing to provide the necessary current. An alternative is to use a piloted valve—a valve which is actuated by pneumatic or hydraulic pressure—and use a low-power solenoid valve to supply the required actuation pressure.

The electrical interface to a solenoid valve depends on its voltage and current requirements. Although AC-powered valves (e.g., 120 VAC) were formerly the norm, the near universality of electronic control has resulted in 24 VDC valves predominating, with lower voltage valves common in some forms of equipment. In any case, solenoid valves are typically controlled through digital outputs of appropriate rating on a programmable logic controller (PLC) or other control system (see 8.2.6: Output Interfacing Techniques).

8.2.3 Proportionate Control Valves

In fluid power applications requiring more exacting control of motion characteristics, proportionate control valves are sometimes employed. These valves are similar in principle to standard solenoid valves, as described above, but are constructed and controlled such that intermediate states between the valve's two extreme positions are possible. The valve can be controlled to slowly meter air or hydraulic fluid to a cylinder, thus allowing more gradual control of the motion, all under electronic control. When used in conjunction with position sensing in a feedback loop, exacting control of position is also possible.

Proportionate control valves are sometimes provided with their own control electronics, and typically require an analog signal (e.g., 0 to ±10 VDC) as a control input.

8.2.4 Electrical Actuation Control

Several forms of electrical actuation are commonly applied in industrial automation today including, of course, AC and DC motors of various sorts and, increasingly, linear motors.

A common motor technology used in undemanding applications is the AC induction motor. These are commonly controlled through the use of *motor starters*, subsystems which typically include a switching mechanism and some form of overload protection. Often, motor starters are equipped with circuitry—typically resistors or inductors—which limit the initial starting current drawn by the motor. Once the motor has begun running, an additional switch is triggered which bridges the limiter, taking it out of the circuit.

Because of the amount of energy used in AC motors, they have been the focus of much innovation and product development in recent years, the details of which would fill many volumes. The common AC induction motor may still be the workhorse of industry, but it has given way in many applications to more advanced technologies.

One of the strongest trends has been toward the more widespread use of variable frequency drives (VFDs) which, instead of driving a motor with a constant 60 Hz (or 50 Hz) AC waveform, use electronic circuitry to vary the frequency of the power being supplied to the motor, thus varying its speed. In applications where the required output from the motor varies greatly and is often less than full output, the VFD can result in considerable savings of energy.

8.2.5 DC Motor Technologies

For many applications in discrete manufacturing, however, much smaller motors are often required, and these are often DC motors. With declining cost in drive technologies and computing power, stepping motors and servo motors are becoming increasingly widespread in use. Motors are covered in more detail in Topic 10, *Motor and Drive Control*, in this book.

The advantages offered by these technologies include the ability to position actuators quickly and accurately under program control, making them ideal for robotics and other applications in which flexibility is paramount. Even the need for linear motion—long a drawback of motors whose motion is inherently rotary—has been addressed through the development of linear motors. These are often patterned after stepping motors, but with the stator stretched out in linear form.

Other advanced and highly specialized forms of electrical actuation are employed in specific industries, such as microelectronics, where unusually demanding requirements are present. Such technologies include piezoconstrictive devices, which make use of microscopic dimensional changes in certain materials when an electrical current is applied, and voicecoil actuators, which make use of the same principle that moves the cone of a loudspeaker.

8.2.6 Output Interfacing Techniques

Virtually all of the actuation devices mentioned in this section will ultimately need to be made subject to electronic control, typically by a programmable controller or other computer-based system. In the case of programmable controllers, several types of output devices are commonly used for this function, typically in the form of output modules to be inserted into the programmable controller system. These output devices will be briefly discussed here.

8.2.7 DC Outputs

Although some industries historically have favored the use of AC-powered devices, low voltage DC devices are typically preferred for new applications, with 24 VDC being a commonly used control voltage. The reasons for this choice include safety, the relaxed wiring standards that are often possible

with low-voltage devices, and the ease and low cost of interfacing such devices to electronic systems. Programmable controller DC output modules containing 16, or even 32 outputs are commonly available, greatly increasing the density of control possible.

The two most common forms of DC outputs are *open-collector transistor* and *field effect transistor (FET)* outputs. The application of these two forms is similar, although the terminology used may differ, so they will be treated identically for most of this discussion.

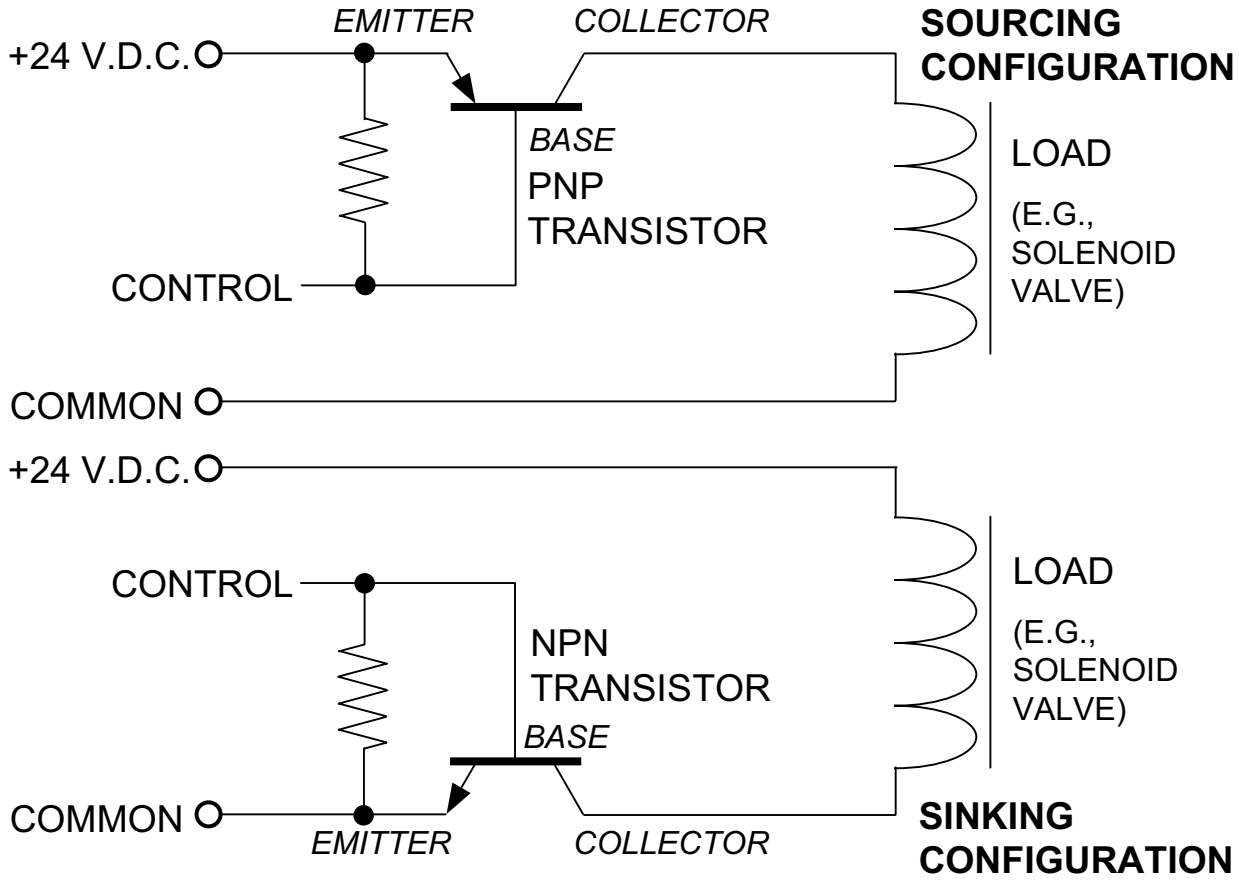


Figure 8-2: Open-Collector Outputs

Open-collector outputs employ transistor devices internally as switches, which are used to switch power to the external load (e.g., solenoid valve, etc.). Transistors are supplied with three terminals, named the *base*, *collector*, and *emitter* (in an FET, they are called the *gate*, *source*, and *drain*). In an open-collector output, the base and emitter are connected to internal circuitry in the output module. The base is the control terminal, often connected to microprocessor circuitry comprising the controller's logic, and the emitter acts as a common terminal, typically connected together with emitters of the other output channels on a given module. The collector terminal is brought out for external connection, hence the term *open-collector*.

8.2.8 Using Open-Collector Outputs

Externally, the load being controlled is connected between the collector and one pole of a power supply, the other pole of the power supply being connected to a *common* terminal on the output module. When a signal is sent from the microprocessor to turn on the transistor, the transistor allows current to flow from the collector to the emitter, essentially acting like a switch which has been turned on. This completes the circuit, sending power to the external load device.

There are two forms of open-collector output modules available—*sourcing* and *sinking*. In the case of sourcing outputs, the common terminal is connected to the positive pole of the DC power supply; the transistors, when switched, will *source* this positive voltage to the load device. In such a case, the other side of each load device is connected to the negative pole of the power supply.

With sinking outputs, the common terminal is connected to the negative pole of the power supply. The transistors, when switched on, will *sink* one terminal of the load device to this negative voltage (often referred to as ground, or common). Assuming the other side of the external load has a fixed connection to the positive pole of the power supply, it will then turn on.

8.2.9 DC Output Precautions

There is a general preference toward the use of sourcing outputs, due to the fact the negative terminal of the power supply is sometimes connected to electrical ground and, therefore, to the frame of an automated system. In the sourcing configuration, the loads have a fixed connection to the negative terminal, and, therefore, an accidental short circuit between a wire leading to a load device and the external metalwork will not cause the unintentional energization of an actuator. With sinking configurations, however, one side of each load device is always connected to a positive voltage source, such that an inadvertent short circuit leading from the other connection on the load device to ground would cause the device to turn on.

In addition, it must be realized that transistors are not perfect switches. When energized, there is a small offset voltage across the transistor, which generates heat equal to the product of current and voltage. This heat determines a maximum current rating for the output, which must be carefully obeyed.

Similarly, transistor outputs have a maximum voltage rating. Usually, the voltage of the DC power supply and control devices used is selected based on this rating. Even so, however, the rating can be inadvertently violated. Inductive devices, such as solenoid coils, motors, or even electromechanical relays, will generate a high voltage when current to the device is interrupted, such as when they are turned off. This voltage, called *back EMF*, can be high enough to destroy a transistor output unless precautionary measures are taken.

Often, a protection diode will be connected in reverse polarity across an inductive load, providing a harmless discharge path for this back-EMF and protecting the output transistor. Some output modules have integrated protection diodes, making this precaution unnecessary.

Another potential source of failure—either of an output transistor or the control system it is part of—is *electrostatic discharge (ESD)* and inducted *electrical noise*. Many processes, particularly high-speed processes involving plastic films, can generate high levels of static electricity, as can human beings walking across a wool carpet in rubber-soled shoes. When the eventual discharge path for this static electricity includes the exposed output or input circuitry of a control system, damage can result. In such environments, voltage limiting devices such as *metal-oxide varistors (MOVs)* may be used to provide some degree of protection, as can shielding of exposed wiring.

Another source of spurious and potentially-damaging high voltage is electrical noise that can be coupled from adjacent wiring. It is common practice to physically separate low-voltage and high-voltage wiring to avoid or limit the amount of inductive or capacitive coupling to sensitive circuits.

An almost universal protection technique used to avoid damage to or malfunction of microprocessor-based control systems is the use of optical isolation between any external connections (inputs or outputs) and the internal microprocessor circuitry. Microprocessors are highly susceptible to even minor fluctuations in voltage, which can cause them to reset or execute randomly, and optoisolation provides an important protective shield between the electrically noisy industrial environment and this sensitive circuitry.

8.2.10 AC Outputs

Electronic control systems employ two different technologies for controlling AC load devices: *triacs* and electromechanical relays. It is assumed that electromechanical relays are well understood—the control system energizes the relay's coil, which electromagnetically switches a mechanical switch, which in turn may be used to control an external load device. The drawbacks of relays are also well known. Over time, the contacts may become pitted and worn, particularly when used to control reactive loads, and will often require replacement well in advance of their solid-state counterparts.

Triacs are semiconductor switches that, when triggered, can pass current in both directions, making them well suited to the control of AC load devices. Some specific characteristics of triacs deserve attention, however, as they can affect the success and reliability of an installation.

When the control signal is removed from a triac, it will remain turned on until such time as the current through it falls to near zero. At this time, the voltage across it must also be near zero or the rapid rise in voltage across the device could spuriously trigger it into conduction again. Unfortunately, when driving an inductive load (such as a solenoid valve, motor, relay, or just about any other device you would typically drive with a triac), the load creates a phase shift between the voltage and current waveforms, such that the triac is unable to turn off. To allow for the triac's proper operation, a *snubber network* composed of a serially connected resistor and capacitor is typically connected across the triac's main terminals, controlling the rate of voltage rise across the device and allowing it to turn off.

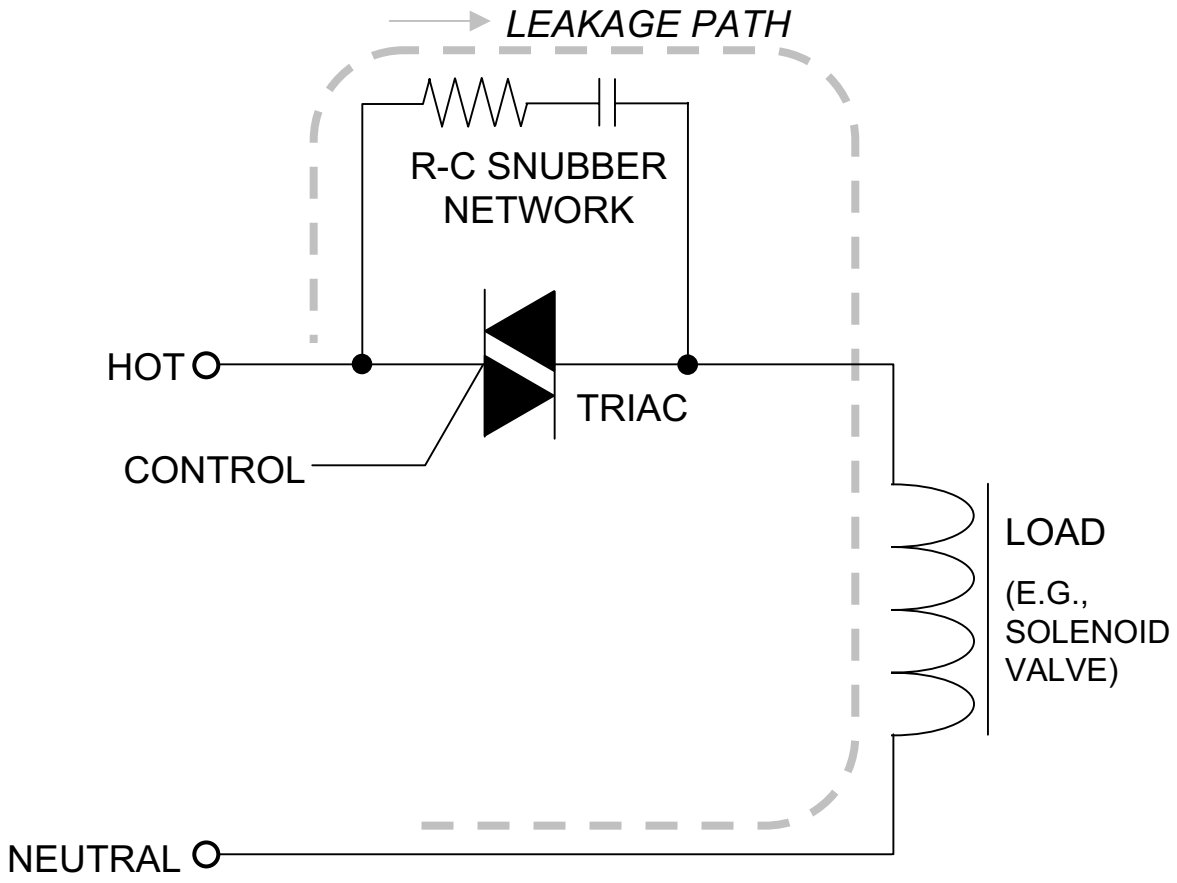


Figure 8-3: Triacs, Snubbers, and the Effects of Leakage

An unfortunate side-effect of this snubber network is that it will always leak a small amount of current. Although small, this current will sometimes be enough to keep a very-low-power device, such as a small solenoid valve, actuated even when the triac output has been turned off. This also means that

a strategy of connecting a triac output to a controller's AC input for signaling purposes will often fail, because the low current requirement of the input will often be met by the leakage current provided by the snubber network, whether the triac output is turned on or not.

Even with this shortcoming, however, triacs and related semiconductor devices are popular and successful choices for AC control, when carefully applied with an understanding of their limitations.

8.3 Sensing Technologies and Interfacing Techniques

Sensing devices have increased dramatically in sophistication as the impetus for their use has evolved over the years. Simple limit switches originally served to detect the completion of a mechanical motion prior to initiation of the next motion, thus preventing damage and allowing a machine to operate at a speed determined solely by its mechanical capabilities.



Figure 8-4: Limit Switches (Courtesy: Omron Electronics LLC)

With the advent of flexible machinery, new sensing technologies such as encoders were employed to allow a single mechanism to accommodate different sizes or configurations of workpieces. Then, as quality issues began to predominate in industrial automation, still more sophisticated technologies such as machine vision systems were deployed to automate high-speed in-process inspection.

This progression represents a gradual supplanting of human involvement in the manufacturing process. Indeed, the earliest "sensors" were the eyes and ears of the operator, who then operated the machine through manual control switches. Although there are certainly a number of areas where electronic sensors exceeded human capabilities—in speed, exactitude, or consistency—some of the more basic judgments made by the human counterpart have taken decades to evolve technology with comparable performance. The simple question, "is the part upside down?" may not be so simple for a sensor to answer.

8.3.1 Limit Switches

In its original meaning, a *limit switch* is a switch placed to sense when a mechanism has reached the end, or limit, of its travel. Although some automated systems are designed to rely on the passage of time alone to infer that a motion has been completed, it is generally considered a safer technique to sense the actual completion of a motion prior to commencing the next part of a sequence of operation.

This is especially true if mechanical damage or danger to an operator could occur in the event of an undetected jam.

The simplest form of limit switch is a mechanical switch designed to be mounted to a machine component. Often, these are equipped with actuators that allow some degree of overtravel without damage to the switch. Depending on the environmental and regulatory requirements of an application, limit switches may be plastic-encased devices with exposed electrical terminals, or fully metal-enclosed devices sealed against moisture.

Driven in large part by reliability concerns, there is a strong trend to the use of non-contact electronic sensors in place of conventional mechanical switches. *Hall-effect sensors*, for example, sense the proximity of a magnetic field, and *proximity switches* are available to sense a variety of materials including nonmetallics. *Photoelectric sensors* send out a beam of light—typically infrared—and sense its reflection off an approaching object. The light beam is often modulated so that ambient light can be effectively filtered from the reflected signal.

Each type of sensor has its applications:

- Some suppliers equip the piston inside a pneumatic or hydraulic cylinder with a magnetic element, allowing Hall-effect sensors to be mounted on the outside of the cylinder to sense the piston at various points in its stroke.
- Proximity sensors can be applied to discriminate between various materials, or to sense objects in environments where optical sensing might be unreliable due to dust, oil or other contaminants.
- Photoelectric sensors can be employed to sense across distances or, using fiber optic technology, in tight quarters where other types of sensing would not be practical.

8.3.2 Interfacing Concerns with Limit Switches

Careful attention must be paid to the electrical characteristics of a limit switch to determine the proper way to interface it to an electronic control system. This is true even with mechanical switches—the low voltages used in some control systems may not be sufficient to break through the layer of oxidation or contamination buildup on mechanical switch contacts, rendering them unreliable. For this reason, gold-plated contacts are frequently used on mechanical switches intended for interfacing to a control system.

Although some sensing devices—notably certain photoelectric sensors—are equipped with internal relays for interfacing, most devices instead have solid-state outputs with limitations that must be obeyed. Two-wire and three-wire devices are commonly available. Two-wire devices rely on a low level of current constantly passing through the device to power its internal electronics. For this reason, the electronic system to which it is connected must allow for this "leakage" current without falsely triggering. When the sensor enters its active state, it draws sufficient additional current to bring the voltage across the sensor down to a few volts—again, the control system must reliably see this state as an *on* condition, even in the presence of this *offset voltage*.

Interfacing three-wire sensors is less exacting, since the three wires leading to the sensor allow for power, a common connection, and a separate output connection from the sensor to the control system. The output is often of the *open-collector* variety and may be sinking or sourcing. Current and voltage ratings must be respected, of course, but the additional concerns of leakage and offset voltage tend not to be significant factors.

8.3.3 Position-Sensing Technologies

There is now frequently a need to obtain position information beyond the simple end-of-stroke indication provided by limit switches. Examples include:

- Feedback confirmation of mechanical positioning (e.g., for robotic arms, X-Y positioning tables, etc.)
- Triggering secondary events at specific points in a primary motion.
- Velocity control feedback, through differentiation of successive position indications.

The most common application for position sensing of this sort is in connection with servo motors (see Topic 11), and the technologies employed for this purpose are incremental and absolute encoders and resolvers.

Two other position sensing technologies are worthy of note, although their application is somewhat more specialized. *Potentiometric sensing* makes use of a linear or rotary *potentiometer*, which consists of an electrically resistive element on which an electrical contact rides. Often, the extreme ends of the resistive element are connected across a precise fixed voltage, at which point the voltage sensed on the moving contact is analogous to its position along the element. This approach provides an analog voltage directly proportionate to position (subject to the linearity of the resistive element). One drawback to be aware of, however, is the relatively limited life of the resistive element due to wear from the moving contact.

Finally, *magnetostrictive sensing* is sometimes applied for linear position sensing of long hydraulic cylinders, for example. This technique uses a long waveguide, down which an electromagnetic reference pulse is induced. A permanent magnet, typically connected to a moving mechanical element (e.g., a piston in a hydraulic cylinder) rides on this waveguide and, when the reference pulse reaches the magnetic field of this magnet a strain pulse is generated which travels back down the waveguide. Precise timing measurement between the induction of the reference pulse and the receipt of the strain pulse provides the position of the magnet, and therefore the piston. Magnetostrictive sensing allows the sensing elements to be completely sealed, making it an appropriate technology for many industrial environments.

8.3.4 Sensing Other Physical Variables

Automation of discrete manufacturing processes today often carries with it the need for sensing a very broad range of physical variables. Often, this is to create a permanent record of manufacturing conditions relating to a given workpiece. Other times, the measurement is used directly as feedback for the process being controlled. Topic 1 in this book discusses pressure and temperature measurements.

8.3.5 Tracking Product Using Electronic Identification

In many cases, automated systems must be able to identify a specific workpiece being acted upon. This is true where customization of individual products to a customer's specification is required, and also where regulatory authorities require tracking and recording of manufacturing data.

One approach to product tracking is the use of barcode technology, through the application of a unique barcode or Data Matrix code to a product or product carrier. Fixed-mount barcode readers are then employed to read and transmit the barcode information to a control system, typically via an RS-232 serial interface.

Another tracking technology that has some advantages for industrial application is RFID *(Radio Frequency IDentification)*. This approach consists of a "tag" capable of transmitting a unique identification number back to a nearby transceiver. *Passive tags* consist simply of an etched antenna and an integrated circuit, and derive sufficient power from the RF energy received by the antenna to power the IC to send a response. *Active tags* contain an internal power source (and are therefore slightly larger) and often have the capability to store additional information received from the transceiver. In environments where a ubiquitous plant network is not feasible, this allows a workpiece equipped with such a tag to accumulate data about its manufacture as it progresses through stages of production.

8.3.6 Machine Vision

Machine vision is a sensing technology which is seeing increasing interest and application due to its potential to extend the reach of automation into operations which formerly required human intervention. Applications for machine vision today include automated inspection for a wide range of characteristics such as placement, dimensional measurements, surface defects and color grading, as well as providing visual feedback for robotic positioning and component orientation.

Available systems span a broad range of cost and capability, from relatively simple and inexpensive systems capable of performing some pattern matching and basic orientation, to *smart cameras* incorporating integrated processors allowing some more advanced functionality, to larger and more complex systems which may be custom programmed for a highly sophisticated tasks. The latter often require the specialized skills of a system integrator experienced in vision system design for successful application.

The electrical interface of a machine vision system involves two aspects. First, a trigger input is typically provided, which may be a signal from a sensor detecting the presence of a workpiece, or a command from a control system. Often, vision systems are provided with high-speed trigger inputs to accommodate machines with high production rates and fast-moving product.

The output from a vision system often takes one of two forms—a simple "good/bad" signal, or more complex data which may include a captured image for subsequent processing or storage. Systems often incorporate discrete outputs which may be programmed to signal, for example, when a stored pattern is matched, and often have additional communications capabilities (serial or Ethernet) to communicate higher-level information to a control system.

8.4 Remote and Networked I/O

A discussion of input/output technologies used in automation control would not be complete without mention of the dramatic shift in I/O architecture taking place today, driven by the prevalence of networking technologies in the plant environment. Where formerly all sensors, valves, and other control devices would be wired back to a single programmable controller, increasingly a more distributed form of connectivity is being applied.

The first step in this progression was the use of "device-level" networks such as Modbus, Profibus and DeviceNet, which provided a means of reducing wiring costs by combining devices on a common serial network. Widespread use of these bus technologies encouraged the development of functional subsystems—for example, motion controllers and temperature controllers—such that time-critical information did not have to be relayed back to a central controller, and low level tasks could be offloaded to the local processing capability of the subsystem. These subsystems would then take their high-level commands from, and provide their high-level results to, the central controller across the bus.

With the near-universal acceptance of Ethernet for plant data communications, variants of each of the device-level serial networking protocols have been developed that may be used across Ethernet. Although the initial impetus for this migration was cost savings—to achieve commonality of equipment and wiring with that already used in the plant—the result has been another rethinking of I/O architecture.

With sensing and actuation no longer tied physically to a proximate control system, there is once more a blurring of actuation and control, such that smart sensors and smart actuators are coming into broad use. Localized processing—in some cases, localized control—has changed and will continue to change the face of sensing and actuation in discrete manufacturing. The functions mentioned in this section will increasingly appear in aggregated form as subsystems on a network, interchanging high-level information with peer and supervisory systems, a most interesting trend to watch.

For more information on networking technology, refer to Topic 22, *Digital Communications*, and Topic 23, *Industrial Networks*.

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